Barret Schloerke

Senior Software Engineer, Posit

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Summary

Senior software engineer for Shiny, Posit's web framework for R and Python (1M+ monthly downloads). My statistician's skepticism (PhD) shapes how I build and test tools that turn data scientists into full-stack app developers. Highlights: built OpenTelemetry tracing into Shiny for R and Python, so users can see where their apps spend time; created shinyreact, so Shiny apps can use React-based UI; and cut Shiny's release testing from 3 weeks for 10 people to under 2 days for 2.

Experience

Senior Software Engineer, Posit · Remote — Jan 2022 – Present

- Core developer of Shiny for R and Python, Posit's flagship framework that brings users to its commercial products.
- Cut CI time 98%, measured by 4+ hours of testing finishing in a 4.5-minute run, by architecting Playwright controllers for every Shiny for Python component and spreading 1000+ tests over 170 jobs (5 Pythons × 3 browsers).
- Brought Shiny for R's best UI to Python by porting bslib's components: sidebars, cards, value boxes, tooltips, and more.
- Built shinylive for R and Python: Shiny apps that run in the browser with no server, even in Quarto sites and slides.
- Enabled every Shiny user to trace apps with OpenTelemetry, build LLM-powered apps (querychat, shinychat), and pair agentic coding with React-based UI (shinyreact).

Software Engineer, RStudio (now Posit) · Remote — Jan 2018 – Dec 2021

- Cut release testing from 3 weeks across 10 people to under 2 days for 2 by architecting shinycoreci: nightly GitHub Actions testing of 150+ apps on Mac, Windows, and Linux × 5 R versions.
- Migrated Shiny's JavaScript to TypeScript, so every PR's client code is type-checked across all files (before, none was).
- Sustained plumber, reactlog, shinytest2, and dozens of other Shiny packages by owning releases, docs, and triage.

Tool Development Team, Gates Foundation: HBGDki · Remote — Apr 2016 – Dec 2018

- Produced custom summary statistics for 75+ datasets, feeding visualization apps that consultants used in work to prevent avoidable infant deaths.

Researcher, DARPA XDATA and D3M grants, Purdue University · West Lafayette, IN — Sep 2016 – Dec 2017

- Delivered a web interface for subject matter experts to explore machine learning pipelines by partnering remotely with Hafen Consulting, Kitware, and KnowledgeVis.

Software Engineer, Metamarkets · San Francisco, CA — Jan 2011 – Jul 2012

- Built interactive D3 visualizations for analytics dashboards, backed by Druid, on DVL, a reactive data visualization library, in CoffeeScript.

Open source software

2,300+ merged pull requests across 170+ GitHub repositories since 2018. Top packages I maintain or author:

`shiny` A web application framework for R and Python · R, TypeScript · Python

`shinyreact` Client-Side 'React' Interface for 'shiny' · R, Python, TypeScript

`shinytest2` Testing for Shiny Applications · R

`plumber` An API Generator for R · R

`promises` Abstractions for Promise-Based Asynchronous Programming · R

`GGally` Extension to 'ggplot2' · R

Education

PhD, Statistics, Purdue University · West Lafayette, IN — 2013 – 2017

BS, Computer Engineering, Iowa State University · Ames, IA — 2006 – 2010

Talks, awards & teaching

45 talks & workshops, 11 videos, and 26 blog posts, listed at schloerke.com

National Science Foundation Graduate Research Fellowship — 2012 – 2017

Google Summer of Code, GGally::ggduo — 2016

Internship Supervisor, Posit · Samuel Bharti: a bioinformatics Shiny showcase, 10 Shiny apps built on 5 packages — 2026

Skills

Languages	R, Python, TypeScript, JavaScript, HTML, CSS, SQL, Bash
Frameworks	Shiny (R and Python), React, plumber, htmltools, ggplot2, tidyverse
Testing & CI	Playwright, pytest, testthat, shinytest2, GitHub Actions
Publishing	CRAN, PyPI, npm, conda-forge, pkgdown, Quarto, GitHub Pages